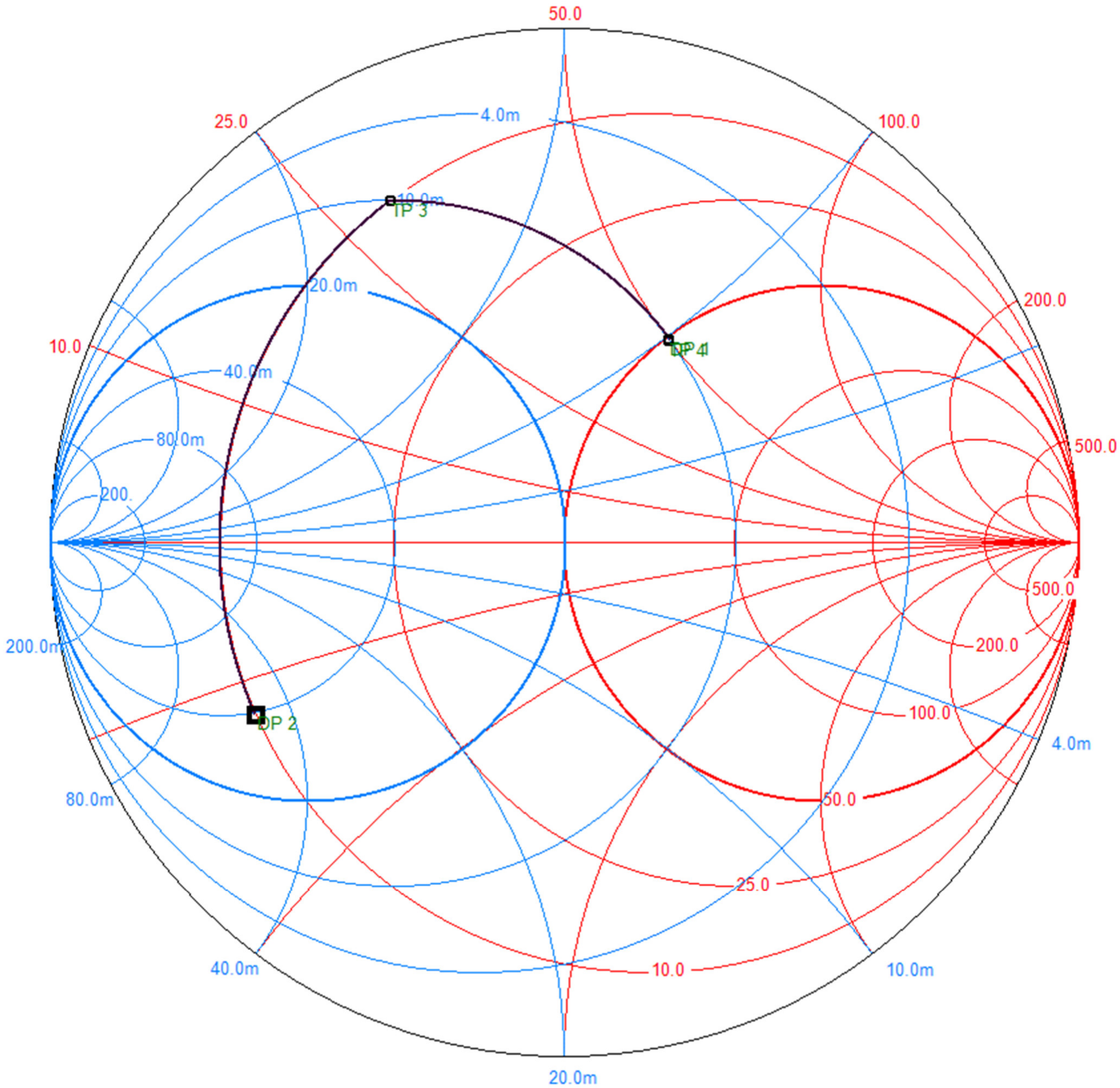
Aufgabe 1: (20 Punkte)

1. $\underline{s}_{11}$ und $\underline{s}_{22}$* direkt ins Smith-Chart einzeichnen oder daraus zunächst $\underline{z}_E$ und $\underline{z}_A$* berechnen:

$$\underline{z}_E = \frac{1+\underline{s}_{11}}{1-\underline{s}_{11}} = \frac{1+0{,}688\cdot e^{-j151^\circ}}{1-0{,}688\cdot e^{-j151^\circ}} = \left(0{,}197 - j0{,}249\right) \qquad \left[\underline{Z}_E = \underline{z}_E \cdot Z_L = \left(9{,}84 - j12{,}46\right)\Omega\right]$$

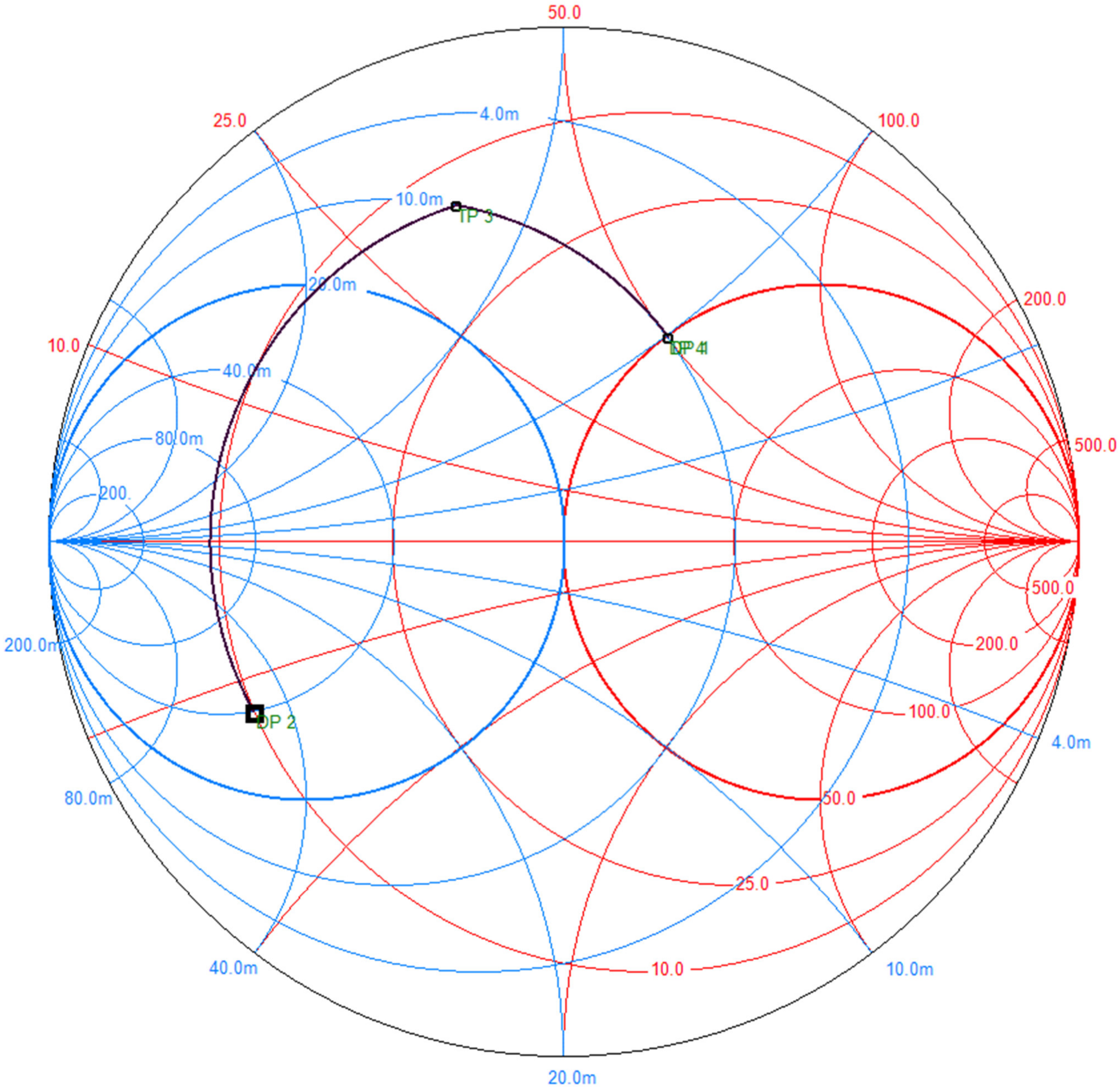
$$\underline{z}_A^* = \left(\frac{1+\underline{s}_{22}}{1-\underline{s}_{22}}\right)^* = \left(\frac{1+0{,}447\cdot e^{-j63^\circ}}{1-0{,}447\cdot e^{-j63^\circ}}\right)^* = \left(1{,}008 + j1{,}003\right) \qquad \left[\underline{Z}_A^* = \underline{z}_A^* \cdot Z_L = \left(50{,}4 + j50{,}2\right)\Omega\right]$$



ablesen: $x_L = 0{,}6 - \left(-0{,}25\right) = 0{,}85 \;\rightarrow\; X_L = x_L \cdot Z_L = 42{,}5\Omega \qquad \underline{\underline{L}} = \frac{X_L}{\omega} = \frac{42{,}5\,\frac{\mathrm{V}}{\mathrm{A}}}{2\pi\cdot 2{,}0\cdot 10^9\,\frac{1}{\mathrm{s}}} = \underline{\underline{3{,}38\mathrm{nH}}}$

$b_C = -0{,}5 - \left(-1{,}5\right) = 1{,}0 \;\rightarrow\; B_C = \frac{b_C}{Z_L} = 20\mathrm{mS} \qquad \underline{\underline{C}} = \frac{B_C}{\omega} = \frac{20\cdot 10^{-3}\,\frac{\mathrm{A}}{\mathrm{V}}}{2\pi\cdot 2{,}0\cdot 10^9\,\frac{1}{\mathrm{s}}} = \underline{\underline{1{,}59\mathrm{pF}}}$

2. Transformation über Leitungen:



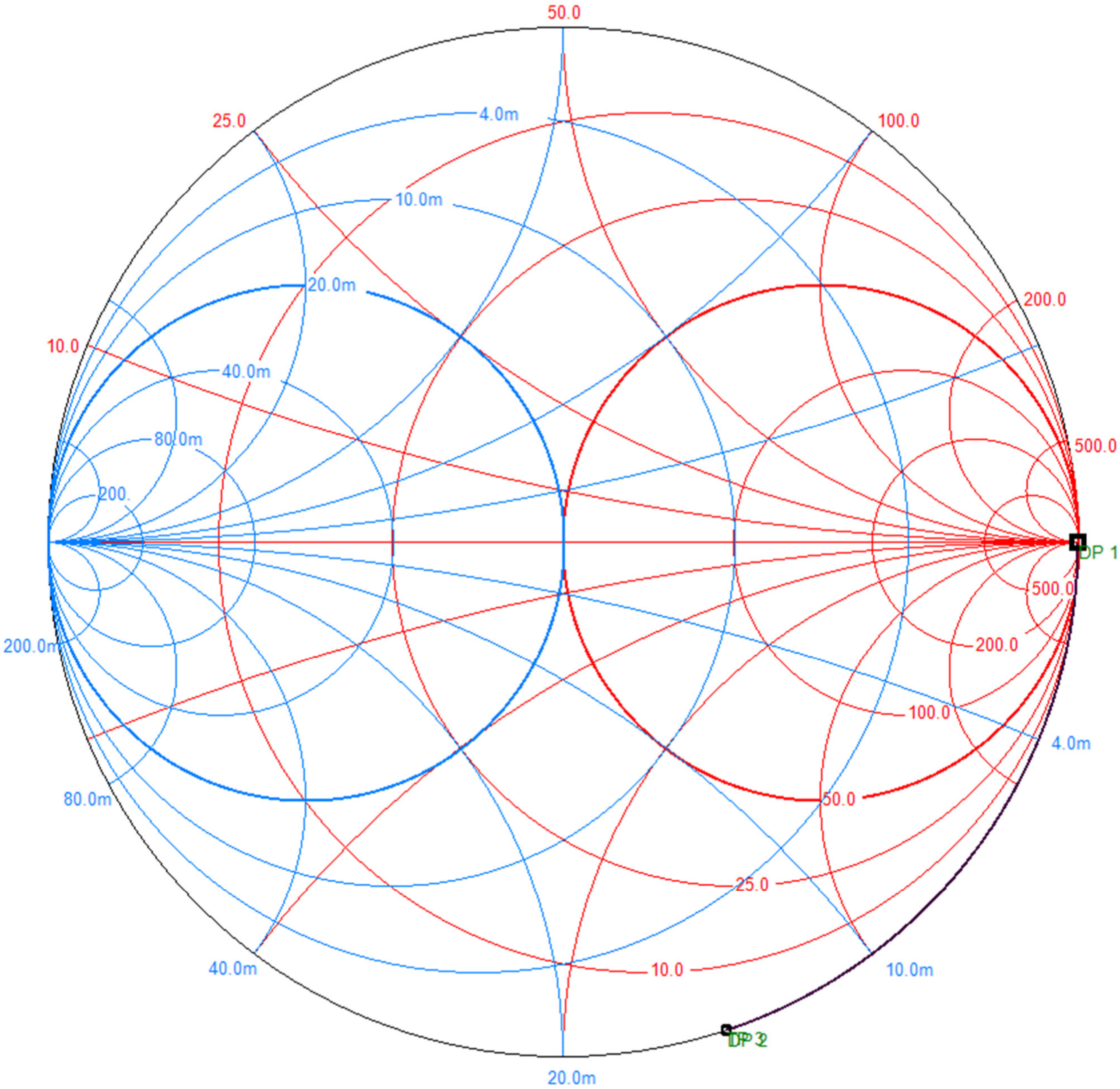
$$\lambda = \frac{c_0}{\sqrt{\varepsilon_r} \cdot f} = \frac{3 \cdot 10^8 \,\frac{\text{m}}{\text{s}}}{\sqrt{2{,}25} \cdot 2 \cdot 10^9 \,\frac{1}{\text{s}}} = 100\text{mm}$$

ablesen:

$$\frac{l_1}{\lambda} = 0{,}101 - 0{,}460 \;\; (+0{,}5) = 0{,}141 \qquad \rightarrow \underline{\underline{l_1}} = 0{,}141 \cdot 100\text{mm} = \underline{\underline{14{,}1\text{mm}}}$$

$$b_c = -0{,}5 - (-1{,}22) = 0{,}72$$

b_c durch Transformation eines Leerlaufs längs einer Leitung darstellen:



$$\frac{l_2}{\lambda} = 0{,}35 - 0{,}25 = 0{,}10 \qquad \rightarrow \; \underline{\underline{l_2}} = 0{,}10 \cdot 100\text{mm} = \underline{\underline{10\text{mm}}}$$

Aufgabe 2: (24 Punkte)

1. $$k_L = \frac{L'_{12}}{L'} = \frac{15\,\frac{\text{nH}}{\text{m}}}{60\,\frac{\text{nH}}{\text{m}}} = 0,25 \qquad k_L = \frac{C'_{12}}{C' + C'_{12}}; \quad k_L \cdot C' + k_L \cdot C'_{12} = C'_{12}; \quad k_L \cdot C' = \left(1 - k_L\right) \cdot C'_{12}$$

$$\underline{\underline{C'_{12}}} = \frac{k_L}{1-k_L} \cdot C' = \frac{0,25}{1-0,25} \cdot 42\,\frac{\text{pF}}{\text{m}} = \underline{\underline{14\,\frac{\text{pF}}{\text{m}}}}$$

2. $$\lambda = \frac{c_0}{f} = \frac{3 \cdot 10^8\,\frac{\text{m}}{\text{s}}}{75 \cdot 10^6\,\frac{1}{\text{s}}} = 400\text{cm}$$

mimimale Koppeldämpfung für $\underline{\underline{l}} = \frac{\lambda}{4} = \underline{\underline{100\text{cm}}}$

damit: $\left|\underline{s}_{41}\right| = \left|\underline{s}_{41}\right|_{min} = k_L \qquad \underline{\underline{C_{min}}} = -20\text{dB} \cdot \lg\left(k_L\right) = -20\text{dB} \cdot \lg\left(0,25\right) = \underline{\underline{12\text{dB}}}$

3. $$\left|\underline{s}_{41}\right|^2 = \frac{k_L^2 \cdot \sin^2 \beta l}{\left(1-k_L^2\right) \cdot \cos^2 \beta l + \sin^2 \beta l} = \frac{k_L^2 \cdot \sin^2 \beta l}{-k_L^2 \cdot \cos^2 \beta l + \cos^2 \beta l + \sin^2 \beta l} =$$

$$= \frac{k_L^2 \cdot \sin^2 \beta l}{1 - k_L^2 \cdot \cos^2 \beta l} = \frac{\left(0,25\right)^2 \cdot \sin^2\left(\frac{2\pi}{400\text{cm}} \cdot 150\text{cm}\right)}{1 - \left(0,25\right)^2 \cdot \cos^2\left(\frac{2\pi}{400\text{cm}} \cdot 150\text{cm}\right)} = 3,226 \cdot 10^{-2}$$

$$\underline{\underline{C}} = -10\text{dB} \cdot \lg\left(\left|\underline{s}_{41}\right|^2\right) = -10\text{dB} \cdot \lg\left(3,226 \cdot 10^{-2}\right) = \underline{\underline{14,9\text{dB}}}$$

4. $$k_L = 10^{-\frac{6\text{dB}}{20\text{dB}}} = 0,50 \qquad Z_L = \sqrt{\frac{L'}{C'}} = \sqrt{\frac{60 \cdot 10^{-9}\,\frac{\text{Vs}}{\text{Am}}}{42 \cdot 10^{-12}\,\frac{\text{As}}{\text{Vm}}}} = 37,8\Omega$$

$$\underline{\underline{Z_{L1}}} = Z_L \cdot \sqrt{1-k_L^2} = 37,8\Omega \cdot \sqrt{1-0,50^2} = \underline{\underline{32,7\Omega}}$$

$$\underline{\underline{Z_{L2}}} = Z_L \cdot \frac{\sqrt{1-k_L^2}}{k_L} = 37,8\Omega \cdot \frac{\sqrt{1-0,50^2}}{0,50} = \underline{\underline{65,5\Omega}}$$

$$\underline{\underline{l_1}} = \underline{\underline{l_2}} = \frac{\lambda}{4} = \underline{\underline{100\text{cm}}}$$

Aufgabe 3: (21 Punkte)

1. zunächst Umrechnung in (Wellenwiderstands-unabhängige) Ein- und Ausgangsimpedanzen:

$$\underline{Z}_e = Z_{L,Bezug} \cdot \frac{1+\underline{s}_{11}}{1-\underline{s}_{11}} = 50\Omega \cdot \frac{1+0,707 \cdot e^{-j45°}}{1-0,707 \cdot e^{-j45°}} = 111,8\Omega \cdot e^{-j63,4°} = (50 - j100)\Omega$$

$$\underline{Z}_a = Z_{L,Bezug} \cdot \frac{1+\underline{s}_{22}}{1-\underline{s}_{22}} = 50\Omega \cdot \frac{1+0,762 \cdot e^{j23°}}{1-0,762 \cdot e^{j23°}} = 205\Omega \cdot e^{j54,8°} = (118 + j167)\Omega$$

$$\underline{s}_{11,75\Omega} = \frac{\underline{Z}_e - \underline{Z}_L}{\underline{Z}_e + \underline{Z}_L} = \frac{(50 - j100)\Omega - 75\Omega}{(50 - j100)\Omega + 75\Omega} = 0,644 \cdot e^{-j65,4°}$$

$$\underline{s}_{22,75\Omega} = \frac{\underline{Z}_a - \underline{Z}_L}{\underline{Z}_a + \underline{Z}_L} = \frac{(118 + j167)\Omega - 75\Omega}{(118 + j167)\Omega + 75\Omega} = 0,677 \cdot e^{j34,7°}$$

2. $$10^{\frac{v_{Tr,max}}{10\text{dB}}} = V_{Tr,max} = \left|\underline{s}_{21,75\Omega}\right|^2 \cdot \frac{1}{1-\left|\underline{s}_{11,75\Omega}\right|^2} \cdot \frac{1}{1-\left|\underline{s}_{22,75\Omega}\right|^2}$$

$$\left|\underline{s}_{21,75\Omega}\right|^2 = 10^{\frac{v_{Tr,max}}{10\text{dB}}} \cdot \left(1-\left|\underline{s}_{11,75\Omega}\right|^2\right) \cdot \left(1-\left|\underline{s}_{22,75\Omega}\right|^2\right) = 10^{\frac{17\text{dB}}{10\text{dB}}} \cdot \left(1-0,50^2\right) \cdot \left(1-0,40^2\right) = 31,57$$

$$\left|\underline{s}_{21,75\Omega}\right| = 5,62$$

3.

ungünstigste Phasenlage für K: $\left|\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}\right|^2$ wird minimal

$$\left|\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}\right|_{min} = \left|\underline{s}_{11}\underline{s}_{22}\right| - \left|\underline{s}_{12}\underline{s}_{21}\right| = 0,707 \cdot 0,762 - 4,0 \cdot 0,015 = 0,4787$$

$$K_{min} = \frac{1-\left|\underline{s}_{11}\right|^2 - \left|\underline{s}_{22}\right|^2 + \left|\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}\right|^2_{\min}}{2\left|\underline{s}_{12}\underline{s}_{21}\right|} =$$

$$= \frac{1-0,707^2 - 0,762^2 + 0,4787^2}{2 \cdot 0,015 \cdot 4,0} = 1,239 > 1 \qquad \text{"stabil"}$$

ungünstigste Phasenlage für B: $\left|\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}\right|^2$ wird maximal

$$\left|\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}\right|_{max} = \left|\underline{s}_{11}\underline{s}_{22}\right| + \left|\underline{s}_{12}\underline{s}_{21}\right| = 0,707 \cdot 0,762 + 4,0 \cdot 0,015 = 0,5987$$

$$B_{min} = 1 + \left|\underline{s}_{11}\right|^2 - \left|\underline{s}_{22}\right|^2 - \left|\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}\right|^2_{max} =$$

$$= 1 + 0,707^2 - 0,762^2 - 0,5987^2 = 0,5607 > 0 \qquad \text{"stabil"}$$

→ Der Transistor ist auf jeden Fall stabil!

Aufgabe 4: (17 Punkte)

1. $p_A + g_{LNB} - a_L + g_{VV} = p_E$

$$\underline{\underline{g_{VV}}} = p_E - p_A - g_{LNB} + a_L = -65\,\text{dBm} - (-91\,\text{dBm}) - 53\,\text{dB} + 46\,\text{dB} = \underline{\underline{19\,\text{dB}}}$$

2.

$$P_{R,Ant} = k \cdot T \cdot B = 1{,}38 \cdot 10^{-23}\,\tfrac{\text{Ws}}{\text{K}} \cdot 35\,\text{K} \cdot 27 \cdot 10^{6}\,\tfrac{1}{\text{s}} = 1{,}3041 \cdot 10^{-14}\,\text{W}$$

$$P_{S,Ant} = 10^{-3}\,\text{W} \cdot 10^{\frac{-91\,\text{dBm}}{10\,\text{dB}}} = 7{,}9433 \cdot 10^{-13}\,\text{W} \quad \rightarrow \quad \left(\frac{S}{N}\right)_{Ant} = \frac{P_{S,Ant}}{P_{R,Ant}} = \frac{7{,}9433 \cdot 10^{-13}\,\text{W}}{1{,}3041 \cdot 10^{-14}\,\text{W}} = 60{,}91$$

$$\left(\frac{S}{N}\right)_{Empf} = 10^{\frac{15\,\text{dB}}{10\,\text{dB}}} = 31{,}6228 \qquad \rightarrow \; F_{ges} = \frac{\left(\frac{S}{N}\right)_{Ant}}{\left(\frac{S}{N}\right)_{Empf}} = \frac{60{,}91}{31{,}6228} = 1{,}9261 \qquad \left[\triangleq 2{,}847\,\text{dB}\right]$$

$$G_{LNB} = 10^{\frac{g_{LNB}}{10\,\text{dB}}} = 10^{\frac{53\,\text{dB}}{10\,\text{dB}}} = 2{,}0 \cdot 10^{5} \qquad F_{VV} = 10^{\frac{f_{VV}}{10\,\text{dB}}} = 10^{\frac{3{,}0\,\text{dB}}{10\,\text{dB}}} = 2{,}0$$

$$G_L = 10^{-\frac{a_L}{10\,\text{dB}}} = 10^{-\frac{46\,\text{dB}}{10\,\text{dB}}} = 2{,}5 \cdot 10^{-5} \qquad F_L = 10^{\frac{a_{L1}}{10\,\text{dB}}} = 10^{\frac{46\,\text{dB}}{10\,\text{dB}}} = 4{,}0 \cdot 10^{4}$$

$$F_{ges} = 1 + (F_{LNB} - 1) + \frac{F_L - 1}{G_{LNB}} + \frac{F_{VV} - 1}{G_{LNB} \cdot G_L}$$

$$F_{LNB} = F_{ges} - \frac{F_L - 1}{G_{LNB}} - \frac{F_{VV} - 1}{G_{LNB} \cdot G_L} = 1{,}9261 - \frac{4{,}0 \cdot 10^{4} - 1}{2{,}0 \cdot 10^{5}} - \frac{2{,}0 - 1}{2{,}0 \cdot 10^{5} \cdot 2{,}5 \cdot 10^{-5}} = 1{,}5261$$

$$\underline{\underline{f_{LNB}}} = 10\,\text{dB} \cdot \lg F_{LNB} = \underline{\underline{1{,}84\,\text{dB}}}$$

<u>Aufgabe 5:</u> (18 Punkte)

1. Reflexionsfaktor der Ersatzquelle:

$$\underline{r}_2 = \underline{s}_{22} + \frac{\underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r}_Q}{1 - \underline{s}_{11} \cdot \underline{r}_Q} = 0,36 + \frac{j0,64 \cdot j0,64 \cdot 0,25 \cdot e^{j30°}}{1 - 0,36 \cdot 0,25 \cdot e^{j30°}} = 0,2668 - j0,0601 = 0,2734 \cdot e^{-j12,7°}$$

$$\underline{\underline{\underline{r}_L}} = \underline{r}_2^* = \underline{\underline{0,2734 \cdot e^{+j12,7°}}}$$

2.

$$\underline{b}_{Q2} = \underline{b}_Q \cdot \frac{\underline{s}_{21}}{1 - \underline{s}_{11} \cdot \underline{r}_Q} = \underline{b}_Q \cdot \frac{j0,64}{1 - 0,36 \cdot 0,25 \cdot e^{j30°}} = \underline{b}_Q \cdot 0,6933 \cdot e^{j92,8°}$$

$$\underline{\underline{P_L}} = \tfrac{1}{2}\left|\underline{b}_{Q2}\right|^2 \cdot \frac{1 - \left|\underline{r}_L\right|^2}{\left|1 - \underline{r}_2 \cdot \underline{r}_L\right|^2} = \tfrac{1}{2}\left|\underline{b}_Q\right|^2 \cdot 0,6933^2 \cdot \frac{1 - 0,28^2}{\left|1 - 0,28 \cdot 0,28\right|^2} = 5,0\,\mathrm{W} \cdot 0,522 = \underline{\underline{2,61\,\mathrm{W}}}$$

3.

$$P_{L,abs,0} = \tfrac{1}{2}\left|\underline{b}_Q\right|^2 \cdot \frac{1 - \left|\underline{r}_L\right|^2}{\left|1 - \underline{r}_Q \cdot \underline{r}_L\right|^2} = 5,0\,\mathrm{W} \cdot \frac{1 - 0,28^2}{\left|1 - 0,25 \cdot e^{j30°} \cdot 0,28\right|^2} = 5,21\,\mathrm{W}$$

$$\underline{\underline{a_E}} = 10\,\mathrm{dB} \cdot \lg\left(\frac{P_{L,abs,0}}{P_L}\right) = 10\,\mathrm{dB} \cdot \lg\left(\frac{5,21\,\mathrm{W}}{2,61\,\mathrm{W}}\right) = \underline{\underline{3,00\,\mathrm{dB}}} \qquad [3,19\,\mathrm{dB}]$$

Reflexionsfaktor der Ersatzlast:

$$\underline{r}_1 = \underline{s}_{11} + \frac{\underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r}_L}{1 - \underline{s}_{22} \cdot \underline{r}_L} = 0,36 + \frac{j0,64 \cdot j0,64 \cdot 0,28}{1 - 0,36 \cdot 0,28} = 0,2325$$

$$P_{Q,ab} = \tfrac{1}{2}\left|\underline{b}_Q\right|^2 \cdot \frac{1 - \left|\underline{r}_1\right|^2}{\left|1 - \underline{r}_Q \cdot \underline{r}_1\right|^2} = 5,0\,\mathrm{W} \cdot \frac{1 - 0,2325^2}{\left|1 - 0,25 \cdot e^{j30°} \cdot 0,2325\right|^2} = 5,24\,\mathrm{W}$$

$$\underline{\underline{a_B}} = 10\,\mathrm{dB} \cdot \lg\left(\frac{P_{Q,ab}}{P_{L,max}}\right) = 10\,\mathrm{dB} \cdot \lg\left(\frac{5,24\,\mathrm{W}}{2,61\,\mathrm{W}}\right) = \underline{\underline{3,03\,\mathrm{dB}}} \qquad [3,21\,\mathrm{dB}]$$